TECHNICAL SUPPLEMENT 03 · ENGLISH

Scenario-Specific Comparisons and Effect Sizes

Agri-Social Navigation in Vineyard Corridors

Reinaldo Betancourt
Ingrid Nicole Vásconez
Viviana Moya
William Chamorro
Sandra Cano
Marco Antonio Molina
Juan Pablo Vásconez

Corresponding author: Juan Pablo Vásconez

Study identity	Specification
Campaign / Campaña	EVAL_20261019_112257
Navigation / Navegación	9 × 4 × 10 = 360; SUCCESS 266; TIMEOUT 94
Benchmark	180; 20 per method / por método
Version / Versión	20 October / octubre 2026

Numerical companion to the revised Sensors manuscript. This report uses the identified endpoint dataset and separates it from the older public campaigns. All seven authors are listed in the original manuscript order.

1. Contrast definition

All contrasts are M4 minus comparator, within scenario. There are 128 records: four endpoints × four scenarios × eight comparators. Two Holm families contain 64 hypotheses each; eight unavailable completed-duration comparisons remain untested placeholders. A negative difference favors M4 for failure, exposure or completed duration, but only within the stated estimand.

Continuous endpoints use Welch mean differences, nominal 95% t intervals and Hedges g; effects include the small-sample correction J = 1 - 3/[4(n1+n2)-9]. Binary endpoints use observed risk differences and Fisher exact tests. Degenerate bootstrap intervals for binary risk differences are not used here as evidence of zero uncertainty; exact event-count intervals appear in Supplement 04.

Failure

E	vs.	n4/nX	M4-X	95% CI	RD / g	p Holm	Status
E1	M0	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	M1	10/10	-1.000	see 04 / ver 04	-1.000	0.0006928	OK
E1	M2	10/10	-1.000	see 04 / ver 04	-1.000	0.0006928	OK
E1	M3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	B1	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	B2	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	B3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	B4	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	M0	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	M1	10/10	-1.000	see 04 / ver 04	-1.000	0.0006928	OK
E2	M2	10/10	-1.000	see 04 / ver 04	-1.000	0.0006928	OK
E2	M3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	B1	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	B2	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	B3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	B4	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	M0	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	M1	10/10	-1.000	see 04 / ver 04	-1.000	0.0006928	OK
E3	M2	10/10	-1.000	see 04 / ver 04	-1.000	0.0006928	OK
E3	M3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	B1	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	B2	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	B3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	B4	10/10	0.000	see 04 / ver 04	0.000	1	OK
E4	M0	10/10	0.000	see 04 / ver 04	0.000	1	OK
E4	M1	10/10	-1.000	see 04 / ver 04	-1.000	0.0006928	OK
E4	M2	10/10	-1.000	see 04 / ver 04	-1.000	0.0006928	OK
E4	M3	10/10	-0.400	see 04 / ver 04	-0.400	1	OK
E4	B1	10/10	-0.500	see 04 / ver 04	-0.500	1	OK
E4	B2	10/10	-0.100	see 04 / ver 04	-0.100	1	OK

E	vs.	n4/nX	M4-X	95% CI	RD / g	p Holm	Status
E4	B3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E4	B4	10/10	-0.400	see 04 / ver 04	-0.400	1	OK

IntimateEvent

E	vs.	n4/nX	M4-X	95% CI	RD / g	p Holm	Status
E1	M0	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	M1	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	M2	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	M3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	B1	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	B2	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	B3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E1	B4	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	M0	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	M1	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	M2	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	M3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	B1	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	B2	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	B3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E2	B4	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	M0	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	M1	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	M2	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	M3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	B1	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	B2	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	B3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E3	B4	10/10	0.000	see 04 / ver 04	0.000	1	OK
E4	M0	10/10	0.000	see 04 / ver 04	0.000	1	OK
E4	M1	10/10	0.000	see 04 / ver 04	0.000	1	OK
E4	M2	10/10	0.000	see 04 / ver 04	0.000	1	OK
E4	M3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E4	B1	10/10	0.000	see 04 / ver 04	0.000	1	OK
E4	B2	10/10	0.000	see 04 / ver 04	0.000	1	OK
E4	B3	10/10	0.000	see 04 / ver 04	0.000	1	OK
E4	B4	10/10	0.000	see 04 / ver 04	0.000	1	OK

Exposure12_s

E	vs.	n4/nX	M4-X	95% CI	RD / g	p Holm	Status
E1	M0	10/10	-28.078	[-32.595, -23.562]	-5.985	4.919e-06	OK
E1	M1	10/10	-148.460	[-149.717, -147.202]	-108.120	<1e-15	OK
E1	M2	10/10	-148.196	[-149.626, -146.766]	-95.738	<1e-15	OK
E1	M3	10/10	1.109	[-0.066, 2.283]	0.860	1	OK
E1	B1	10/10	3.032	[1.693, 4.370]	2.082	0.006229	OK
E1	B2	10/10	-0.269	[-2.368, 1.830]	-0.121	1	OK
E1	B3	10/10	2.200	[0.776, 3.623]	1.427	0.1056	OK
E1	B4	10/10	7.152	[6.480, 7.823]	10.326	6.952e-08	OK
E2	M0	10/10	-9.845	[-10.013, -9.678]	-55.620	<1e-15	OK
E2	M1	10/10	-140.348	[-140.514, -140.182]	-809.904	<1e-15	OK
E2	M2	10/10	-140.355	[-140.521, -140.190]	-812.778	<1e-15	OK
E2	M3	10/10	4.627	[4.432, 4.823]	21.366	<1e-15	OK
E2	B1	10/10	1.074	[-1.613, 3.761]	0.387	1	OK
E2	B2	10/10	-6.977	[-7.155, -6.799]	-35.971	<1e-15	OK
E2	B3	10/10	3.414	[3.228, 3.599]	16.765	2.09e-15	OK
E2	B4	10/10	7.314	[7.137, 7.491]	37.972	<1e-15	OK
E3	M0	10/10	-25.769	[-26.302, -25.237]	-43.541	<1e-15	OK
E3	M1	10/10	-136.522	[-136.928, -136.116]	-324.829	<1e-15	OK
E3	M2	10/10	-136.682	[-137.099, -136.264]	-307.585	<1e-15	OK
E3	M3	10/10	2.291	[1.588, 2.995]	2.968	0.0001259	OK
E3	B1	10/10	0.540	[-4.804, 5.884]	0.098	1	OK
E3	B2	10/10	-8.701	[-12.089, -5.314]	-2.478	0.00592	OK
E3	B3	10/10	1.991	[1.585, 2.398]	4.726	4.035e-05	OK
E3	B4	10/10	6.490	[6.084, 6.896]	15.448	1.505e-09	OK
E4	M0	10/10	-5.732	[-8.068, -3.395]	-2.222	0.002287	OK
E4	M1	10/10	-130.053	[-132.095, -128.011]	-60.735	<1e-15	OK
E4	M2	10/10	-129.301	[-131.329, -127.272]	-61.266	1.854e-15	OK
E4	M3	10/10	4.832	[0.729, 8.935]	1.084	0.4376	OK
E4	B1	10/10	-5.701	[-19.024, 7.621]	-0.412	1	OK
E4	B2	10/10	-20.790	[-40.235, -1.345]	-1.033	0.657	OK
E4	B3	10/10	5.838	[2.723, 8.954]	1.695	0.02181	OK
E4	B4	10/10	11.079	[7.597, 14.560]	2.898	0.0001583	OK

DurationCompleted_s

E	vs.	n4/nX	M4-X	95% CI	RD / g	p Holm	Status
E1	M0	10/10	14.592	[12.672, 16.512]	7.364	1.295e-06	OK
E1	M1	10/0	--	--	--	--	NE
E1	M2	10/0	--	--	--	--	NE
E1	M3	10/10	-11.438	[-15.979, -6.897]	-2.342	0.003304	OK

E	vs.	n4/nX	M4-X	95% CI	RD / g	p Holm	Status
E1	B1	10/10	3.601	[0.891, 6.311]	1.197	0.2299	OK
E1	B2	10/10	11.173	[9.248, 13.098]	5.600	9.863e-06	OK
E1	B3	10/10	10.970	[8.881, 13.059]	4.816	6.983e-07	OK
E1	B4	10/10	10.391	[8.435, 12.347]	5.038	8.036e-06	OK
E2	M0	10/10	14.740	[14.435, 15.046]	46.585	5.864e-14	OK
E2	M1	10/0	--	--	--	--	NE
E2	M2	10/0	--	--	--	--	NE
E2	M3	10/10	-2.729	[-3.440, -2.018]	-3.567	5.797e-05	OK
E2	B1	10/10	-1.539	[-5.147, 2.069]	-0.412	1	OK
E2	B2	10/10	11.812	[11.505, 12.118]	37.035	2.051e-13	OK
E2	B3	10/10	10.971	[10.646, 11.296]	31.231	<1e-15	OK
E2	B4	10/10	5.871	[4.628, 7.114]	4.502	3.03e-05	OK
E3	M0	10/10	14.490	[14.097, 14.884]	35.641	8.455e-13	OK
E3	M1	10/0	--	--	--	--	NE
E3	M2	10/0	--	--	--	--	NE
E3	M3	10/10	-5.620	[-8.202, -3.038]	-2.095	0.01654	OK
E3	B1	10/10	-0.530	[-5.414, 4.355]	-0.105	1	OK
E3	B2	10/10	8.220	[7.691, 8.750]	13.969	1.008e-15	OK
E3	B3	10/10	10.488	[10.095, 10.881]	25.795	1.491e-11	OK
E3	B4	10/10	9.340	[8.945, 9.736]	22.657	1.473e-11	OK
E4	M0	10/10	43.859	[41.427, 46.290]	17.476	6.61e-10	OK
E4	M1	10/0	--	--	--	--	NE
E4	M2	10/0	--	--	--	--	NE
E4	M3	10/6	-24.573	[-31.745, -17.401]	-4.735	0.003043	OK
E4	B1	10/5	-8.771	[-36.455, 18.912]	-0.649	1	OK
E4	B2	10/9	34.220	[31.472, 36.969]	11.351	1.364e-12	OK
E4	B3	10/10	29.949	[23.458, 36.440]	4.320	1.424e-05	OK
E4	B4	10/6	17.209	[10.799, 23.618]	3.583	0.009181	OK

NE = not estimable. Nominal intervals do not imply simultaneous 95% family coverage. Failure contrasts concern observed mission outcomes; completed-duration contrasts concern completers only.

Source and scope

Source: data/; results/expected/. PDFs provide the reading reports; CSV/JSON retain numerical inputs and provenance. Scope: evaluated simulation, without new experiments, independent raw-stream reconstruction or physical validation.